NAMAA Community at AlexandriaX-2026: Prompting, Fine-Tuning and Agreement Voting for Dialectal Arabic Translation and Evaluation

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Abstract

Arabic encompasses diverse spoken varieties with limited parallel data, making both translation and automatic evaluation difficult beyond Modern Standard Arabic. We present the NAMAA Community submissions to all three AlexandriaX-2026 subtasks: English-to-dialect dialogue translation, cross-dialect translation, and dialectal MT error detection. Our systems combine few-shot prompting, fine-tuned encoder-decoder and decoder-only models, and encoder- and LLM-based span detection. Held-out ablations show that minimum-Bayes-risk consensus can suffer from model-family imbalance; we therefore use per-dialect routing for translation and 2-of-3 character-level agreement for error-span detection. Our systems rank fourth and fifth of six in the unconstrained and constrained Subtask 1 tracks, third of four on Subtask 2, and second of six on Subtask 3. The Subtask 3 system achieves 0.5448 overlap-F1, exceeding the overall winner on span localization, with category assignment remaining the main weakness. Code and resources are publicly available.[1]

1 Introduction

Arabic comprises diverse spoken varieties that differ substantially from Modern Standard Arabic (MSA) and from one another, making dialectal translation and evaluation challenging. AlexandriaX-2026 (El Mekki et al., 2026) addresses these challenges through three subtasks spanning thirteen Arabic dialects and English: context-aware English-to-dialect dialogue translation based on the Alexandria dataset (EL Mekki et al., 2026) (Subtask 1), banking-domain cross-dialect translation following the AraFinNLP setting (Malaysha et al., 2024) (Subtask 2), and localization and categorization of dialectal MT errors using LQM (Magdy et al., 2026) (Subtask 3). We present the NAMAA Community systems for all three subtasks and analyze the design choices underlying their performance.

Our systems combine prompted frontier models, fine-tuned encoder–decoder and decoder-only translators—including the dialect-focused NileChat (El Mekki et al., 2025)—and span encoders, with combination strategies selected empirically. For Subtask 1, held-out experiments show that minimum-Bayes-risk (MBR) consensus (Kumar and Byrne, 2004; Eikema and Aziz, 2020; Freitag et al., 2022) can be degraded by model-family imbalance, motivating per-dialect routing. For Subtask 3, we derive a character-level overlap-F1 break-even condition that motivates agreement-based span selection.

Our contributions are threefold: (1) systems for all AlexandriaX-2026 subtasks combining fine-tuning, prompting, routing, and span agreement; (2) a held-out analysis of consensus failure under model-family imbalance; and (3) a metric-derived criterion for precision-oriented span combination in dialectal MT error detection.

2 Related work

Research on dialectal Arabic spans three closely related strands. The first concerns corpus construction and variety identification, with resources such as MADAR (Bouamor et al., 2018), the NADI shared-task series (Abdul-Mageed et al., 2021b), QADI (Abdelali et al., 2021), and IADD (Zahir, 2022) providing data and benchmarks for modelling variation across Arabic varieties. A second strand focuses on generation and translation. Multilingual MT systems such as NLLB-200 (NLLB Team et al., 2022) expose dedicated target codes for several Arabic varieties, while models such as AraT5 (Nagoudi et al., 2022) and NileChat (El Mekki et al., 2025)

[1] https://huggingface.co/collections/NAMAA-Space/namaa-at-alexandriax-2026

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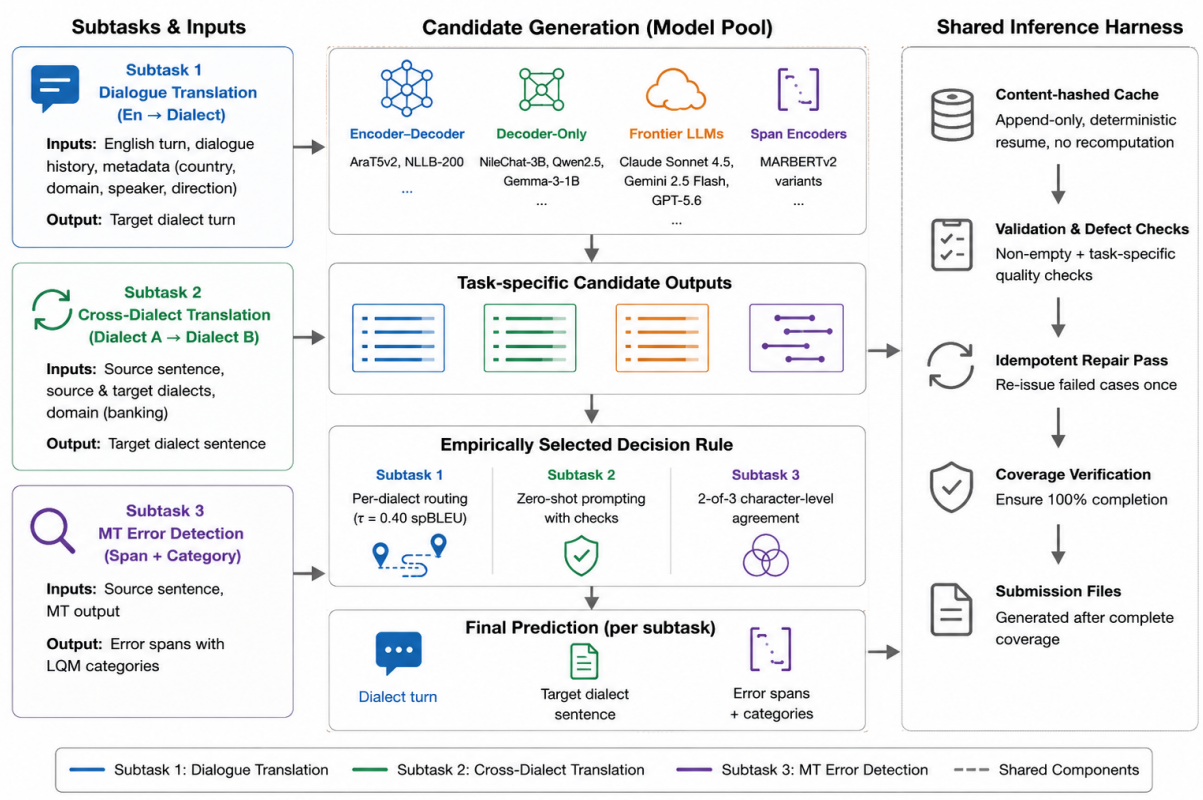


Figure 1: Overview of the NAMAA Community systems for AlexandriaX-2026.

pursue Arabic- and dialect-focused pre-training. Prompted large language models further provide competitive translation quality when supplied with suitable in-context exemplars (Brown et al., 2020; Vilar et al., 2023; Liu et al., 2022; Agrawal et al., 2023).

A third strand concerns evaluation. MQM-style frameworks (Lommel et al., 2014; Freitag et al., 2021) represent translation quality through localized and categorized errors, while Jarrar et al. (2023) introduce the banking-domain *ArBanking77* corpus underlying the domain setting extended by Subtask 2. Less attention has been paid to how heterogeneous dialectal-MT systems should be combined once their outputs are available. Although minimum-Bayes-risk decoding is commonly reported as beneficial (Freitag et al., 2022), our held-out experiments show that, for the candidate pool studied here, model-family imbalance can make consensus underperform the strongest individual system. We therefore treat system combination as an empirical problem, evaluating both routing for translation and a metric-derived break-even criterion for span-level ensembling.

3 Task Setup and System Overview

Figure 1 summarizes the submission pipeline. AlexandriaX-2026 covers three dialectal Arabic translation and evaluation tasks. Subtask 1 translates English dialogue turns into Arabic dialects using conversational and speaker metadata, with constrained ($\leq$3B) and unconstrained tracks; it provides 66,480 training turns, 12,250 development turns over eleven dialects, and a 14,459-turn blind test over thirteen dialects, including test-only Libyan and Sudanese.

Subtask 2 performs banking-domain cross-dialect rewriting over 3,574 blind-test instances spanning 30 directed variety pairs, with no parallel training or development split. Subtask 3 detects and categorizes MT error spans, with 1,125/138/145 train/dev/test instances over five directions. Full task statistics and examples appear in Appendix B.

Shared inference harness. All systems write to an append-only, content-hashed cache that supports deterministic resume and avoids recomputation. Outputs are accepted only if non-empty and valid; failures pass through an idempotent repair stage before final coverage checks. This recovered all but one of 535 Subtask 1 test turns (3.7%) that ini-

tially returned empty outputs, largely from apparent safety false positives on agricultural wording.

Subtask 1: dialogue translation. The constrained track fine-tunes AraT5v2, NLLB-200, NileChat-3B, Qwen2.5-1.5B (Qwen Team, 2024), and Gemma-3-1B (Gemma Team, 2025), with AraT5v2 as the ranked submission. The unconstrained pool adds Claude Sonnet 4.5 and Gemini 2.5 Flash using five dialect- and domain-matched exemplars,[2] plus a beam-decoded Gemma fine-tune of ≈3.1B parameters, a larger checkpoint than the Gemma-3-1B constrained candidate, which is the ranked entry. That checkpoint was initially entered in the constrained track; because ≈3.1B exceeds that track's ≤3B cap it is officially recorded under the unconstrained track, and AraT5v2 is therefore our ranked constrained entry.

Held-out experiments show that MBR selection by mean chrF agreement underperforms the strongest individual system because multiple related models can reinforce one another. We therefore use per-dialect routing: a dialect switches from the globally strongest system only when another candidate improves by more than $\tau = 0.40$ spBLEU, reducing sensitivity to noise in ≈1,100-turn per-dialect estimates. The margin was fixed a priori as a noise guard and was not tuned over candidate values. The formal rule appears in Appendix C.

Subtask 2: cross-dialect translation. With no parallel train/dev split, Gemini 2.5 Flash is used zero-shot with explicit source and target varieties and instructions to preserve meaning, avoid MSA drift, and prevent unnecessary copying. Few-shot retrieval is impractical because 3,518 of 3,574 test rows have distinct sources. Outputs are checked for truncation, script anomalies, and inappropriate identity mappings, with failures routed through the shared repair stage.

Subtask 3: MT error detection. Span localization and category assignment are separated because they favor different models. Candidate spans come from a MARBERTv2 encoder union U (Abdul-Mageed et al., 2021a), GPT-5.6 (G), and Gemini 2.5 Flash (*Gem*), while GPT supplies the labels. Since additional characters improve overlap-F1 only when their marginal precision exceeds half the current score (Appendix C), the submitted system keeps only characters supported by at least two sources:

$$S = (U \cap G) \cup (U \cap Gem) \cup (G \cap Gem).$$

Offsets are reconstructed by string matching, yielding exact recovery on all 2,240 gold spans used for validation.

4 Experimental Setup

Splits and evaluation. Supervised systems are fine-tuned on the official training conversations. Ablations (5) are fitted and evaluated on disjoint halves of the 12,250-turn development set, except for the submitted routing table, which uses scored blind-test feedback (Appendix F); the resulting selection bias is discussed in the Conclusion. Few-shot exemplars are drawn only from conversations outside the development split. Subtasks 1 and 2 are ranked by spBLEU, with chrF++ as the secondary metric (Goyal et al., 2022; Post, 2018; Popović, 2015, 2017), both macro-averaged over countries. The published Subtask 3 documentation defines Overall using exact-match F1 and class-macro-F1; however, the official scored submissions are reproduced to six decimal places by averaging character-level overlap-F1 and class-macro-F1. We therefore report the empirically observed scorer behavior; formal definitions are given in Appendix A. Adapters use LoRA (Hu et al., 2022) with 4-bit quantization (Dettmers et al., 2023); encoder-decoder systems use beam search, and the remaining fine-tuned systems use greedy decoding. Hyperparameters, prompts, auxiliary-corpus construction, and library versions are reported in Appendices K–N.

5 Results and Analysis

Overall results. Table 1 summarizes the four ranked entries. In Subtask 1, beam-decoded Gemma and the routed ensemble are effectively tied (27.413 vs. 27.412 spBLEU), while routing leads by 0.47 chrF++. For Subtask 3, our second-ranked system achieves 0.5448 overlap-F1, exceeding the winner's 0.5426; the ranking gap instead comes from category assignment (0.3834 vs. 0.4538 class-macro-F1). Similarly, our Subtask 2 system exceeds the second-ranked team on chrF++ despite a 1.3-point spBLEU deficit.

Architecture and combination. The unconstrained Subtask 1 entry improves over our constrained system by 4.16 spBLEU, yet scale alone does not explain performance: the 220M encoder–decoder AraT5v2 outperforms larger decoder-only

[2] Accessed via API as `claude-sonnet-4.5`, `gemini-2.5-flash`, and, for Subtask 3, `gpt-5.6`, July 2026.

Subtask / track	System	Development		Blind test		Rank
		Avg spBLEU *S3: Overall*	**Avg chrF++** *S3: overlap-F1*	**Avg spBLEU** *S3: Overall*	**Avg chrF++** *S3: overlap-F1*	
S1 constrained ($\leq$3B)	**AraT5v2, 220M, enc–dec (submitted)**	**25.13**	**40.66**	**23.26**	**39.03**	**5/6**
	Qwen2.5-1.5B LoRA, dec-only	23.71	40.51	21.24	38.06	
	NileChat-3B QLoRA, dec-only	23.54	39.68	—	—	
	Gemma-3-1B LoRA, dec-only	22.71	38.86	20.09	36.20	
	NLLB-200 LoRA, enc–dec	21.83	38.13	—	—	
S1 unconstrained	Claude Sonnet 4.5, 5-shot	—	—	26.36	42.26	
	Gemini 2.5 Flash, 5-shot	—	—	26.68	42.49	
	Routed ensemble	—	—	27.412	**43.05**	
	Gemma, $\approx$3.1B, beam search (ranked)	—	—	**27.413**	42.58	**4/6**
S2 cross-dialect	**Gemini 2.5 Flash, zero-shot (submitted)**	—	—	**24.53**	**39.04**	**3/4**
S3 error detection	**2-of-3 agreement (submitted)**	—	0.5126‡	**0.4641**	**0.5448**	**2/6**

Table 1: Principal systems and official results. ‡2-of-2 intersection, the development analogue of the submitted system; its class term was not evaluated on development, so only overlap-F1 is shown.

models, while 1.3B NLLB-200 is weakest despite sharing spBLEU's SentencePiece vocabulary. This suggests that adaptation to the target distribution can matter more than parameter count under the constraint (Appendix H).

Combination is likewise not automatically beneficial (Appendix D). Per-dialect routing yields only $+0.07$–$+0.28$ spBLEU, whereas MBR consensus loses 0.42 and a top-2-per-dialect vote loses 0.80. The candidate pool contains several related NileChat variants that reinforce one another and can outvote a stronger individual system, making consensus sensitive to model-family multiplicity. The 32.25 spBLEU per-turn oracle nevertheless shows that strong outputs are often already present, leaving selection as a major bottleneck.

Subtask 3 and broader implications. Across five scored Subtask 3 submissions, Overall rises from 0.3367 to 0.4641 (Appendix G), with the largest gain coming from agreement-based span selection rather than increased coverage. This matches the break-even analysis in Appendix C: added characters help only when their marginal precision is sufficiently high. Taken together, the results favor measurement-driven combination over default ensembling: task adaptation can outweigh scale in translation, while precision-oriented agreement can outperform broader coverage in error detection.

Qualitative analysis. The errors support these trends. In Subtask 1, references often preserve dialect-specific code-switching that systems normalize into Arabic; for example, Moroccan gold may retain *le certificat* while the system produces الشهادة. Latin-script material appears in 40.0% of Moroccan development references but only 0.8% of Syrian ones (Appendix I). In Subtask 2, errors mainly involve over-editing closely related varieties whose correct outputs may remain unchanged. In Subtask 3, remaining errors concentrate in adjacent LQM categories, such as *pragmatics* versus *sociolinguistics*, consistent with the class-macro-F1 deficit. Additional examples appear in Appendix O.

6 Conclusion

This paper presents the NAMAA Community systems for all three AlexandriaX-2026 subtasks. Our experiments show that system combination must be validated rather than assumed to help: MBR consensus and voting can underperform the strongest individual system under model-family imbalance, while Subtask 3 benefits from agreement-based span selection guided by a metric-derived break-even condition. These findings are limited by the use of scored blind-test feedback for some routing and subtask 3 decisions, the relatively small model pool used for the MBR analysis, and the absence of human or neural-metric evaluation. Despite these constraints, the results support measurement-driven combination strategies for dialectal Arabic translation and evaluation.

Limitations

Blind-test feedback in system selection. Our strongest claims rest on held-out development evidence, but two deployment decisions do not. The routing strategies compared in Appendix D are fitted and evaluated on disjoint halves of the development conversations, so the comparison between routing, MBR consensus, and the per-turn oracle is held out rather than leaderboard-driven. The *specific* routing table submitted for Subtask 1 (Ap-

pendix F), by contrast, is fitted on measured per-dialect blind-test scores, and the Subtask 3 configuration was chosen across five scored submissions (Appendix G). Those two artefacts therefore carry test-set selection bias and should be read as leaderboard-tuned engineering, not as generalizable findings. The routing margin $\tau = 0.40$ spBLEU was fixed a priori as a noise guard for $\approx$1,100-turn per-dialect estimates and was never swept over candidate values, so it does not itself contribute selection bias.

Subtask 1 track attribution. Our two Subtask 1 entries were initially submitted under the wrong tracks because of a coordination error within the team, and the mismatch was found only after blind-test scoring. The $\approx$3.1B beam-decoded Gemma checkpoint was entered in the constrained track, and the routed ensemble in the unconstrained track. Since $\approx$3.1B parameters exceed the constrained track's $\leq$3B cap, the Gemma checkpoint is not constrained-eligible, and the official standings record it under the unconstrained track (rank 4/6 at 27.413 spBLEU) with the AraT5v2 fine-tune as our constrained entry (rank 5/6 at 23.257). We report those official attributions throughout. The routed ensemble, which we had intended as the unconstrained entry, scores 27.412 and is therefore effectively tied with the entry that was actually ranked; we report it as a contrastive system rather than as a ranked result. Readers comparing our constrained result against other teams should note that it comes from a 220M-parameter model, the strongest of our $\leq$3B candidates on the blind test.

The agreement rule is not independently validated. The break-even condition in Eq. (6) is analytic, and the region decomposition in Appendix J bounds what the unscored combinations in that family could have achieved. Neither establishes that 2-of-3 is the best available threshold. We did not compare agreement thresholds on a strictly held-out development split, so the benefit we report may be partly specific to the blind-test selection process rather than a robust property of agreement-based span combination.

Metric ambiguity in Subtask 3. The published task documentation defines Overall using exact-match F1, whereas the official scored submissions are reproduced by averaging character-level overlap-F1 with class-macro-F1. We report the empirically observed scorer behaviour and flag the discrepancy rather than resolving it; absolute Subtask 3 numbers are therefore comparable within this shared task but not necessarily across papers that assume the documented definition.

Small and correlated candidate pool. The MBR analysis uses five systems, several of which are NileChat variants differing only in context and auxiliary data. This is precisely the regime in which consensus tracks model family rather than quality, so our negative result for MBR should not be read as evidence that MBR fails in general. Whether consensus recovers under a pool with substantially greater architectural and pre-training diversity remains untested here.

Encoder–decoder scale. Our best constrained system is a 220M-parameter AraT5v2 fine-tune, and the encoder–decoder comparison tops out at 1.3B with NLLB-200. We did not evaluate larger Arabic-specialized encoder–decoder models, so our claim that task adaptation can outweigh scale is established only within this range and under the shared task's parameter constraint.

Surface-overlap evaluation only. All results use spBLEU and chrF++, which reward n-gram and character overlap with a single reference. Our own error analysis shows this penalizes valid dialectal renderings that differ lexically from the annotators' conventions. We report no human judgements and no neural metrics, so the rankings reflect reference fidelity rather than adequacy as such.

No development split for Subtask 2. Subtask 2 releases only a blind-test set. Our system is therefore zero-shot by necessity, with no reference-based validation of prompt or decoding choices before submission.

Ethics Statement

This work uses the official AlexandriaX-2026 shared-task data together with publicly available dialectal Arabic corpora, listed in Appendix N. We collected no new data from human participants and processed no personally identifying information; ingestion of speech corpora reads transcript columns only.

Translation quality is markedly uneven across varieties: Mauritanian Hassaniya is our weakest dialect by a wide margin. Systems that appear serviceable in aggregate can therefore fail disproportionately for speakers of under-represented varieties,

and macro-averaged scores conceal this. We report per-dialect results throughout so that this asymmetry stays visible. None of these systems is suitable for deployment in settings where a mistranslation carries material consequence, such as medical, legal, or financial advice, and our banking-domain results in particular should not be read as evidence of production readiness.

Part of our pipeline queries proprietary models through commercial APIs, which sends shared-task inputs to third-party providers. All such inputs are public shared-task material. We also observed apparent safety false positives on benign agricultural wording, which silently suppressed output for a small fraction of turns; practitioners relying on hosted models for dialectal data should expect and check for this failure mode rather than assume full coverage.

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A Metric Definitions

For Subtasks 1 and 2, let p_n denote modified n-gram precision and BP the brevity penalty. spBLEU is computed as

$$\text{spBLEU} = \text{BP} \cdot \exp\left(\frac{1}{4}\sum_{n=1}^{4} \log p_n\right). \quad (1)$$

For chrF++, let P and R denote precision and recall over character and word ℓ-grams. The corresponding F-score is

$$\text{chrF}_\beta = \frac{(1+\beta^2)\,\text{P}\,\text{R}}{\beta^2\text{P} + \text{R}}, \quad (2)$$

with $\beta = 2$, character ℓ-grams up to order 6, and word ℓ-grams up to order 2.

For Subtask 3, let P and G denote the predicted and gold character sets, respectively. Character-level overlap-F1 is

$$F_1^{\text{ovl}} = \frac{2|P \cap G|}{|P| + |G|}. \quad (3)$$

The official Overall score averages overlap-F1 with macro-F1 over the LQM categories $\mathcal{C}$ appearing in either the gold or predicted output:

$$\text{Overall} = \frac{1}{2}\left(F_1^{\text{ovl}} + \frac{1}{|\mathcal{C}|}\sum_{c\in\mathcal{C}} F_1^{c}\right). \quad (4)$$

B Task Formats and Data

Table 2 summarizes the released data. Subtask 1 provides training, development, and blind-test dialogue turns; Subtask 2 provides only the blind-test cross-dialect set; and Subtask 3 provides train, development, and test instances for span-level MT error detection.

The expected input–output format for each subtask is:

	Input	Output
S1	EN turn + conversation, country, domain, speaker, direction	one dialectal Arabic turn
S2	Arabic sentence + source–target variety pair	sentence rewritten in the target variety
S3	source + MT output	error spans: {character start, end, LQM category}

	Train	Dev	Test	Coverage
S1	66,480	12,250	14,459	13 dialects, 11 domains
S2	—	—	3,574	30 directed pairs, banking
S3	1,125	138	145	5 directions, 6 LQM labels

Table 2: Data statistics by subtask. Subtask 2 provides no parallel training or development split. Subtask 1 development covers eleven of the thirteen test dialects; Libyan and Sudanese are test-only.

Representative examples from the official data and submitted systems are shown below.

S1 — Egyptian, commerce

in *Good morning, how can I help you?*
out صباح الخير، أقدر أساعدك إزاي؟

S2 — Lebanese → Saudi

in شو الخيارات المتاحة عندي بالنسبة للدفع؟
out وش الخيارات المتاحة عندي للدفع؟

S3 — ENG_MAU

in لا حول ولا قوة إلا بالله، نطلبو الله العافية
out نطلبو [26:31] → `sociolinguistics`

Subtask 1 covers thirteen dialects: Egyptian, Jordanian, Lebanese, Libyan, Moroccan, Mauritanian, Omani, Palestinian, Saudi, Sudanese, Syrian, Tunisian, and Yemeni. Its eleven domains are agriculture, commerce, construction, education, energy, everyday/social, healthcare, legal/financial, logistics, professional, and tourism.

Subtask 3 covers English translation into Egyptian, Mauritanian, Moroccan, Palestinian, and Emirati. Its LQM inventory contains {`sociolinguistics`, `semantics`, `morphosyntax`, `orthography`, `pragmatics`, `graphetics`}. The 243 gold development spans are distributed as 58% sociolinguistics, 25% semantics, 9% morphosyntax, 5% orthography, and 3% pragmatics. This concentration in the first two categories limits the class-macro-F1 attainable by systems that predict only a restricted subset of labels.

C Combination Rules

This section formalizes the two combination rules used in 3.

Per-dialect routing (Subtask 1). Let $s_m(c)$ denote the measured spBLEU of system m on dialect c, let m_0 be the globally strongest system, and define

$$\delta(c) = \max_m s_m(c) - s_{m_0}(c).$$

The selected system is

$$m^*(c) = \begin{cases} \arg\max_m s_m(c), & \delta(c) > \tau, \\ m_0, & \text{otherwise}, \end{cases} \quad (5)$$

with $\tau = 0.40$ spBLEU. The margin reduces sensitivity to small differences in per-dialect scores, which are estimated from approximately 1,100 turns per dialect.

Span break-even condition (Subtask 3). Because overlap-F1 in Eq. (3) is computed over characters, every new predicted character increases the denominator, while only genuine error characters increase the numerator. Suppose an additional source contributes d_{pred} new predicted characters, of which d_{tp} are true positives. Adding that source improves overlap-F1 precisely when

$$\frac{d_{tp}}{d_{\text{pred}}} > \frac{F_1^{\text{ovl}}}{2}. \quad (6)$$

Thus, newly added coverage is beneficial only when its marginal hit rate exceeds half of the current overlap-F1; below this threshold, additional coverage reduces the score.

D Combination Ablation

Table below compares Subtask 1 combination strategies. Each strategy is fitted on one half of the development conversations and evaluated on the disjoint half, so these measurements are held out rather than oracle. Δ denotes the reported change relative to the best single system.

Strategy (held-out dev)	Avg spBLEU	Δ
Best single system (AraT5v2)	24.98	—
Route per country	25.06	+0.08
Route per country + sub-dialect	**25.26**	**+0.28**
Route per country + domain	24.66	−0.32
MBR consensus, 5 systems	24.56	−0.42
MBR consensus, top-2 per dialect	24.18	−0.80
Per-turn oracle (upper bound)	*32.25*	+7.27

Two additional strategies perform substantially worse: sampling-based MBR within a single model reaches 20.09 spBLEU, while linear adapter merging reaches 19.90, both well below the strongest individual system.

E Official Standings

The following table reports the final blind-test standing for every track entered. spBLEU is the primary metric for Subtasks 1 and 2, while Overall is the primary metric for Subtask 3. chrF++ and overlap-F1 are the corresponding secondary metrics. **Best** gives the winning team's primary score; bold secondary scores indicate cases where NAMAA exceeds the team ranked immediately above it.

Track	Rank	Primary	Secondary	Best
S1 unconstrained	4/6	27.413	42.584	33.489
S1 constrained (≤3B)	5/6	23.257	39.034	30.423
S2 cross-dialect	3/4	24.528	**39.036**	28.401
S3 error detection	2/6	0.4641	**0.5448**	0.4982

F Subtask 1 Routing Table

The table below gives blind-test spBLEU by dialect for the four unconstrained routing candidates. Bold indicates the candidate selected by Eq. (5). Here C denotes Claude 5-shot, G5 Gemini 5-shot, G0 Gemini zero-shot, and A AraT5v2, which is also the constrained submission.

	C	G5	G0	A
EG	**34.00**	32.52	32.84	26.82
JO	**32.51**	32.05	31.53	29.93
LB	**29.03**	27.92	27.53	26.28
LY	21.91	25.18	**25.64**	15.62
MA	23.38	**23.33**	22.66	18.46
MR	10.08	11.24	9.92	**15.46**
OM	22.04	22.52	21.37	**23.12**
PS	29.46	**30.09**	29.69	26.27
SA	**31.39**	30.84	30.02	26.14
SD	24.37	**25.85**	26.18	15.66
SY	**35.17**	34.52	31.21	33.97
TN	27.17	**27.91**	23.96	24.71
YE	22.22	**22.86**	21.48	19.90
avg	26.36	26.68	25.70	23.26

For MA and SD, Gemini 5-shot is retained despite a nominally higher competing score because the corresponding margins (0.05 and 0.33) remain below $\tau = 0.40$. Averaging the selected per-dialect scores predicts a macro spBLEU of 27.412; the scored submission obtains 27.4121.

G Subtask 3 Submission Progression

The five scored blind-test submissions show how span and category decisions affected the final Overall score.

Span set (blind test)	ovl-F1	cls-MF1	Overall
MARBERTv2 + GPT classes	0.4133	0.2600	0.3367
Encoder union	0.4775	0.2927	0.3851
Union + GPT spans	0.4609	0.3251	0.3930
Per-direction mix	0.5149	0.3312	0.4230
2-of-3 agreement	**0.5448**	**0.3834**	**0.4641**

Increasing coverage is beneficial only while Eq. (6) is satisfied. Extending the encoder union with GPT spans clears the break-even criterion in four of the

five directions but fails for ENG_MAU, where the marginal precision of GPT-only coverage falls below the required threshold. Reverting this direction yields a predicted overlap-F1 of 0.5149, closely matching the scored value of 0.514871.

The third row illustrates a different trade-off. Adding GPT spans globally reduces overlap-F1 from 0.4775 to 0.4609, yet raises Overall because the gain in class-macro-F1 outweighs the span loss. The subsequent per-direction mixture recovers the overlap degradation by reverting ENG_MAU alone. Examples of the resulting span predictions, including one false positive and two category confusions, appear in Appendix O.

H Development-Set System Scores

The following systems are evaluated on the full 12,250-turn Subtask 1 development set covering eleven countries. Parameter counts and architectures are included for the fine-tuned systems discussed in 5.

System	Params/arch.	spBLEU	chrF++
AraT5v2 fine-tune	220M, enc–dec	25.13	40.66
Qwen2.5-1.5B LoRA	1.5B, dec-only	23.71	40.51
NileChat-3B, context-free	3B, dec-only	23.54	39.68
NileChat-3B, +context	3B, dec-only	22.87	39.11
NileChat-3B, +context+aux	3B, dec-only	22.77	38.71
Gemma-3-1B LoRA	1B, dec-only	22.71	38.86
NLLB-200 LoRA	1.3B, enc–dec	21.83	38.13
MBR over 3 NileChat variants	—	23.57	39.86
MBR over 5 samples, one model	—	20.09	37.50
Linear adapter merge	—	19.90	35.33

Parameter count does not predict performance. The 220M encoder–decoder AraT5v2 is strongest, the 1.3B encoder–decoder NLLB-200 is weakest, and among decoder-only systems the ordering Qwen2.5-1.5B > NileChat-3B > Gemma-3-1B runs counter to parameter size. One interpretation is that spBLEU rewards fidelity to the annotators' dialectal conventions more strongly than generic generative fluency. A translator adapted directly to the released targets can acquire these conventions, whereas a larger decoder-only model may produce fluent but lexically different outputs.

For example, for the Egyptian-agriculture sentence *"God willing, it will be a profitable day for everyone"*, AraT5v2 closely reproduces the reference's formulaic ربنا يرزقنا جميعا, whereas decoder-only systems produce fluent but more literal variants resembling إن شاء الله يبقى يوم كويس للكل. Both are valid Arabic renderings, but the latter receives lower n-gram overlap. Under the parameter constraint, task adaptation therefore provides greater returns than scale alone; above the constraint, scale and in-context exemplars become more competitive.

I Code-Switching Rates

Table below reports the proportion of development turns containing at least one Latin-script word. Code-switching is strongly dialect-dependent, and all three systems under-produce it in most listed dialects; AraT5v2 on Tunisian is the exception.

% turns, Latin script	MA	TN	LB	MR	SY
Gold reference	40.0	25.8	19.2	5.3	0.8
AraT5v2 (constrained)	35.6	36.2	14.7	1.5	1.0
NileChat-3B	7.9	24.5	7.6	0.5	1.0
NLLB-200	10.5	18.4	3.2	4.8	1.0

Across all eleven development dialects, gold references contain code-switching in 9.9% of turns (18.0 Latin words per 100 turns), while AraT5v2 reaches 9.6% (14.8 words per 100 turns). The aggregate rates therefore appear similar, but the per-dialect distribution exposes substantial mismatches.

J Subtask 3 Region Decomposition

The three span sources partition predicted characters into seven disjoint regions: U only, 691; G only, 1217; *Gem* only, 1457; $U \cap G$, 234; $U \cap$ *Gem*, 485; $G \cap$ *Gem*, 785; and $U \cap G \cap$ *Gem*, 1326.

Three true-positive totals are known from scored probes: $tp(U) = 1172$, $tp(U \cup G) = 1590$, and $tp(\text{2-of-3}) = 1358$. Subtracting these constraints leaves one free parameter,

$$u = tp(U \text{ only}) \in [0, 232].$$

Every unscored combination in this family can therefore be expressed as an exact function of u and is bounded above by approximately 0.471 Overall. In particular, adding the GPT-only region cannot outperform the submitted combination: it would require a hit rate of 0.276, whereas its maximum feasible rate is 0.191.

K Instruction Prompt for Subtask 1

The prompted Subtask 1 systems use the following template:

```
You are an expert translator. Translate
the English sentence into {dialect}.

   ### Metadata:
- Country: {country}
- Domain: {domain}
- Participants: {participants}
- Speaker: {speaker}
```

```
- Speaker Direction: {direction}

   ### Conversation History:
{history}

   ### Sentence to Translate:
{source}

   ### Translation:
```

In the context-free setting, the history block is replaced by a fixed "no previous turns" marker. For the NileChat fine-tune, the context-free variant reaches 23.54 spBLEU compared with 22.87 when conversation history is included. Context and metadata are therefore omitted for that configuration because their inclusion measurably reduces development performance.

L Hyperparameters

For LoRA-based systems, the adapter rank is 16, $\alpha = 32$, and dropout is 0.05. NileChat-3B, Qwen2.5-1.5B, and Gemma-3-1B target the `q`, `k`, `v`, `o`, `gate`, `up`, and `down` projections and are trained using each model's native chat template. NLLB targets its attention and feed-forward projections. Qwen and Gemma are trained on 8×A100-80GB GPUs.

Training uses 4-bit NF4 quantisation with double quantisation, `bfloat16` compute, paged 8-bit AdamW, a cosine learning-rate schedule with warmup, learning rate 2×10^{-4}, effective batch size 32, and gradient checkpointing with `use_reentrant=False`. Labels are masked with -100. Training runs for one epoch (987 steps).

The implementation uses `transformers` 5.8 (Wolf et al., 2020), `peft` 0.19 (Mangrulkar et al., 2022), `trl` 1.6 (von Werra et al., 2020), and `sacrebleu` 2.6 (Post, 2018).

M NLLB Target Codes

The NLLB targets are: Egyptian `arz_Arab`; Moroccan `ary_Arab`; Tunisian `aeb_Arab`; Saudi `ars_Arab`; Syrian and Lebanese `apc_Arab`; Jordanian and Palestinian `ajp_Arab`; and Yemeni `acq_Arab`.

Libyan, Sudanese, Omani, and Mauritanian have no native NLLB target code and are mapped to the nearest available variety: `arz`, `arz`, `ars`, and `ary`, respectively. Because spBLEU uses NLLB's SentencePiece model for tokenisation, the NLLB system and evaluation metric share the same subword vocabulary.

N Auxiliary Corpus

An auxiliary corpus of approximately 1.8×10^5 cleaned dialectal sentences was assembled from public sources including IADD (Zahir, 2022), QADI (Abdelali et al., 2021), NADI-2021 (Abdul-Mageed et al., 2021b), MADAR (Bouamor et al., 2018), and Egyptian, Saudi, Moroccan, Levantine, Libyan, Sudanese, and Yemeni web and transcript collections.

Ingestion is audio-free: for speech datasets, only transcript columns are read through column projection. URLs, mentions, and emoji are removed, examples with an Arabic-character ratio below 0.5 are discarded, and near-duplicates are filtered. In hindsight, this Arabic-ratio filter is in tension with the code-switching analysis in Appendix I, because it preferentially removes the heavily code-switched Maghrebi material that appears in the gold references.

The corpus is additionally back-translated with NLLB (Sennrich et al., 2016), yielding approximately 3×10^5 synthetic pairs. This augmentation slightly lowers development performance (24.36 versus 24.84 spBLEU), so no submitted system uses the synthetic pairs.

O Qualitative Examples and Errors

This section presents representative outputs from the submitted systems. Subtask 1 examples come from the development set and therefore include gold references; Subtasks 2 and 3 use blind-test examples for which public gold references are unavailable.

For Subtask 3, one quantitative observation motivates the examples below. MARBERTv2 predicts only two of the six LQM categories, which caps its class-macro-F1 at 0.3746 irrespective of boundary accuracy. GPT categories reach 0.3837 when evaluated on gold span boundaries. Combined with GPT's lower span precision, this asymmetry motivates separating span localisation from category assignment (3).

Subtask 1 — correct dialectal output. The following Moroccan agriculture example uses واش as the interrogative particle and dialectal lexical choices rather than MSA.

> EN *The season here is almost finished. Have you found any work for afterwards?*
> SYS الموسم هنا قرب يسالي. واش لقيتي شي خدمة من بعد؟

Subtask 1 — Mauritanian Hassaniya. Mauritanian is the weakest Subtask 1 dialect at 15.5 sp-BLEU. Its orthographic and lexical conventions are challenging for models trained predominantly on Mashriqi varieties; in the example below the system also drops the French term *carburant* retained by the reference.

EN *Full tank of diesel, please. The wait is worth it.*
SYS تانصيوه من گزوال ايجازيك. اتحران ازين.
GOLD أمل كونتينيرا من tnarubrac . تستحق تحاو.

Subtask 1 — code-switching failure. In Moroccan healthcare, the system translates a French lexical item into Arabic although the gold reference preserves the code-switch.

EN *He told me to ask you for the certificate today.*
SYS قال ليا نطلب منك الشهادة ليوم.
GOLD هو لي قاليا ناخد tacfiitrec el من عندك ليوما.

The same pattern occurs at phrase level in a Moroccan logistics example, where the reference preserves a larger French expression.

EN *The latest Logistics Performance Index data is in. Where are the biggest challenges according to the report?*
SYS آخر المعطيات ديال مؤشرات الأداء اللوجستية وصلت. وين أكبر الصعوبات ديالنا حسب التقرير؟
GOLD آخر بيانات ديال ل ecnamrofrep ed ecidni جاو. euqitsigol فين كاينين المشاكل الكبيرة من ل troppar؟

Subtask 1 — idiomatic under-translation. Both outputs below are semantically valid, but only the reference uses the formulaic Egyptian expression.

EN *God willing, it will be a profitable day for everyone.*
SYS إن شاء الله يبقى يوم كويس للكل.
GOLD ربنا يرزقنا جميعا.

Subtask 1 — lexical drift in Moroccan. The system output remains interpretable but is substantially more literal than the idiomatic reference.

EN *And the moment you find one, a parking attendant materializes out of thin air!*
SYS او ملي كتلقى واحد، كيولي ل gnikrap كيعطي هواء نقي!
GOLD او حتا فاش كتلقا بلاسة ، كيجي عندك لعساس مكتعرفو منين خرج .

Subtask 2 — cross-dialect rewriting. For closely related varieties, many transformations are small but systematic, affecting interrogatives, clitics, or aspectual forms.

Lebanese → Saudi شو الخيارات المتاحة عندي
وش الخيارات المتاحة عندي للدفع؟ ⇒ بالنسبة للدفع؟

Egyptian → Moroccan عملية السحب لسه معلقة،
واش عملية السحب باقا معلقة، علاش؟ ⇒ ليه؟

Saudi → Lebanese فيه مبلغ على حسابي، بس أنا ما
في مبلغ بحسابي، بس أنا ما استعملتو ⇒ استخدمته

The first pair replaces شو with وش, while the third changes the object clitic ـه→ـو. When two varieties genuinely coincide, the system leaves the source unchanged: this occurs for 47 rows (1.3%), including 26 Lebanese↔Palestinian examples. Such identity outputs are treated as valid rather than forcing unnecessary edits.

Subtask 3 — tagged spans. The dominant predicted category is *sociolinguistics*: an output may remain intelligible while using a form that is inappropriate for the target variety.

ENG_MAU لا حول ولا قوة إلا بالله، نطلبو الله العافية
SPAN نطلبو [26:31] → `sociolinguistics`

ENG_MAU مائة ألف، الحمد لله، والله جبرت خاطري.
SPAN مائة [0:4] → `sociolinguistics`

ENG_PAL إنت مثال للشباب، الشجاعة، الحب، والحيوية، بدون أي مقدمة.
SPAN بدون أي مقدمة [42:55] → `semantics`

مائة is valid MSA but is marked as inappropriate for the Hassaniya target variety. The third span represents a semantic error: the literal rendering does not preserve the intended idiomatic meaning.

Subtask 3 — system errors on development data. The final examples illustrate one false positive and two category confusions.

ENG_PAL إن شاء الله، من خلال هالأعمال، أنا بخطط إنو أجعل أم الحيتان سنغافورة جديدة.
SYS من خلال → `sociolinguistics`
GOLD أجعل (`sociolinguistics`), الحيتان (`orthography`)

ENG_MOR span خدمتينيش — predicted `semantics`, gold `morphosyntax`

ENG_PAL span ما، — predicted `pragmatics`, gold `sociolinguistics`

P Packaging and Scoring Notes

Several implementation details affected submission validity rather than model quality.

Subtask 3 archive format. The scorer searches for a prediction *JSONL* file. The baseline notebook instead packages a `.json` array, which produces `No prediction JSONL found`. A valid archive must therefore contain exactly one `submission.jsonl` file at its root; placing the file inside an additional wrapper directory also prevents discovery.

Archive filename length. CodaBench rejects uploads whose zip filename exceeds 64 characters, independently of the contents of the archive.

Subtask 2 encoding. Submission CSVs must be written without a byte-order mark (BOM). Otherwise, readers that do not explicitly request `utf-8-sig` may interpret the first header name incorrectly, breaking organiser-side row alignment.

Subtask 3 class-macro-F1. The class term averages over categories appearing in either the gold or predicted annotations. Predicting a category absent from gold therefore introduces an additional zero-valued class term. On development data, two spurious `graphetics` predictions reduce Overall by 0.028.